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$$f(x, y) = e^{(x-2y^2)^2}$$

Eerste orde:

$$\frac{\partial f}{\partial x} = e^{(x-2y^2)^2} \cdot (2x - 4y^2)$$

$$\frac{\partial f}{\partial y} = e^{(x-2y^2)^2} \cdot (-8xy + 16y^3)$$

Tweede orde:

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} &= e^{(x-2y^2)^2} \cdot (2x - 4y^2)^2 + 2e^{(x-2y^2)^2} \\ &= 2e^{(x-2y^2)^2} \cdot (2(x - 2y^2)^2 + 1)\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial y^2} &= e^{(x-2y^2)^2} \cdot (-8xy + 16y^3)^2 + e^{(x-2y^2)^2} \cdot (-8xy + 48y^2) \\ &= 8e^{(x-2y^2)^2} \cdot (8(-xy + 2y^3)^2 - x + 6y^2)\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial x \partial y} &= e^{(x-2y^2)^2} \cdot (2x - 4y^2)^2 + e^{(x-2y^2)^2} \cdot (-8y) \\ &= 4e^{(x-2y^2)^2} \cdot ((x - 2y^2)^2 - 2y)\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial y \partial x} &= e^{(x-2y^2)^2} \cdot (-8xy + 16y^3)^2 + e^{(x-2y^2)^2} \cdot (-8y) \\ &= 8e^{(x-2y^2)^2} \cdot (8(-xy - 2y^3)^2 - y)\end{aligned}$$

References